

up to cyclic permutations as 1-zero is the imposition of the $n-3$ constraints

$$c_{13} = c_{14} = \cdots = c_{n,n-1} = 0$$

on the kinematic data.

Since $c_{ij} = X_{ij} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}$,
the constraints can be equivalently written as

$$\left\{ \begin{array}{l} X_{13} + X_{24} - X_{14} - X_{23} = 0 \\ X_{14} + X_{25} - X_{15} - X_{24} = 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - X_{1n} - X_{2,n-1} = 0 \end{array} \right.$$

$\Rightarrow$

$$\left\{ \begin{array}{l} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2n} = X_{1n} - X_{13} = -X_{13} \end{array} \right.$$

so the 1-zero fully fixes X_{2j} once X_{1j} is specified, and does not affect X_{ij} if $i, j \notin \{1, 2\}$.

The corresponding BCFW shift is the charge

$$\left\{ \begin{array}{l} P_2 \rightarrow P_2 + zq \\ P_n \rightarrow P_n - zq \end{array} \right.$$

where z is a complex parameter and q satisfies

$$\left\{ \begin{array}{l} q^2 = 0 \\ q \cdot P_2 = q \cdot P_n = 0 \end{array} \right.$$

so that

$$(P_2 + zq)^2 = 0, \quad (P_n - zq)^2 = 0.$$

Let $X_{ij}(z)$ be the z -dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly one of P_2, P_n in the sum, otherwise either there is no z or they cancel out.

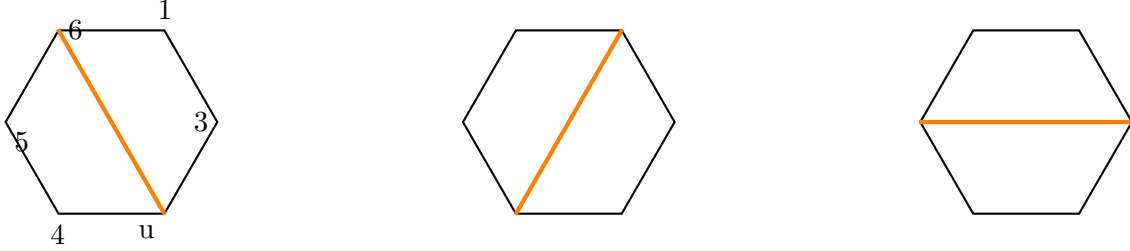
$\Rightarrow$ the only z -dependent ones are $X_{1j}(z)$ $j = 3, \dots, n-1$

and $X_{2j}(z)$ $j = 4, \dots, n$.

Let X_{ij}^∞ denote the coefficient of the largest power of z appearing in $X_{ij}(z)$.

All possible diagrams with 6 legs

group them by



strictly depend on z scaling of projectors

Boundary projectors: the ones that scale linearly with z under BCFW shift (X_{1j} and X_{2j})

group by # of boundary projectors $\Rightarrow$ group them by scaling under BCFW shift

$$\mathcal{B} = \mathcal{B}_0 + \mathcal{B}_{-1} + \mathcal{B}_{-2} + \cdots$$

(with 0, 1, 2, ... boundary projectors)